Ease Patch: Sanofi 2025 Device Innovation Challenge

Year	May – October 2025
Image	
Brief summary	EasePatch was developed for the Sanofi 2025 Device Innovation Challenge to reimagine large-volume on-body drug delivery systems (OBDS). The concept focuses on comfort, discretion, and patient confidence, transforming drug delivery into a lifestyle-integrated experience. EasePatch combines a smart adhesive patch with an optional fashion-forward belt that offers flexibility, safety, and inclusivity while supporting sustainability and regulatory readiness.
Role	Project Manager, UX Research, Product Design, and Human Factors Integration

Project Overview

EasePatch was created for the Sanofi 2025 Device Innovation Challenge to reimagine large-volume on-body drug delivery systems (OBDS).

The goal was to design a connected, comfortable, and sustainable device that improves patient confidence, usability, and adherence to self-injection treatments.

“Care that’s easy to wear.”

EasePatch integrates patient-centered design, engineering, and empathy to make drug delivery effortless and dignified.

Project Goal

To design a patient-first OBDS that enhances confidence and comfort during self-injection while aligning with sustainability and regulatory standards.

Objectives

- Increase user trust and reduce anxiety through clear feedback systems
- Create an adaptable, discreet wearable for diverse users
- Use sustainable, biocompatible materials suitable for long-term production
- Integrate digital connectivity for real-time feedback and adherence tracking

Research & Discovery

Context

Chronic disease patients often face emotional and physical discomfort during drug delivery. Existing OBDS devices focus on function over empathy. EasePatch aims to bridge this gap between usability and emotional design.

Research Highlights

- 1 in 3 self-injecting patients report anxiety during application
- Key pain points: discomfort, visibility, adhesive irritation, and uncertainty
- Market trend: growing demand for connected health wearables and biofeedback-driven designs

User Interviews

Interviews with 10 participants revealed three recurring needs:

- Reduced pain and skin irritation
- Confidence through feedback and easy guidance
- Discreet, lightweight form for daily wear

Personas

Meet the patient personas

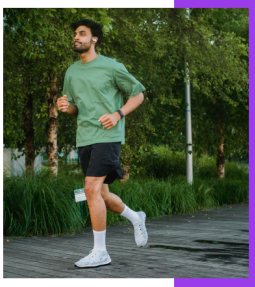
Maya

48 years old
Wife, and mother of two children
High school teacher
Avid hiker
Diagnosed with HER2-positive breast cancer

After several IV infusions at the hospital, Maya is starting self-subcutaneous injections into the thigh, using a fixed-dose formulation to be administered over 8 minutes.

Meet the patient personas

Alex

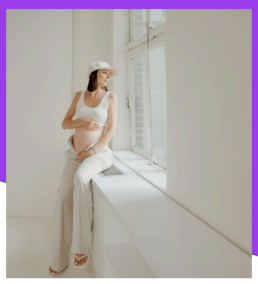


31 years old
Marketing Manager & Marathon Runner
High Tech Adopter
Type 1 Diabetes

Alex is a competitive amateur marathoner who was diagnosed with T1D in their early 20s. They are data-driven, tech-savvy, and constantly optimizing their performance, both in their career and their running. Their diabetes management is a key part of their training regimen.

Meet the patient personas

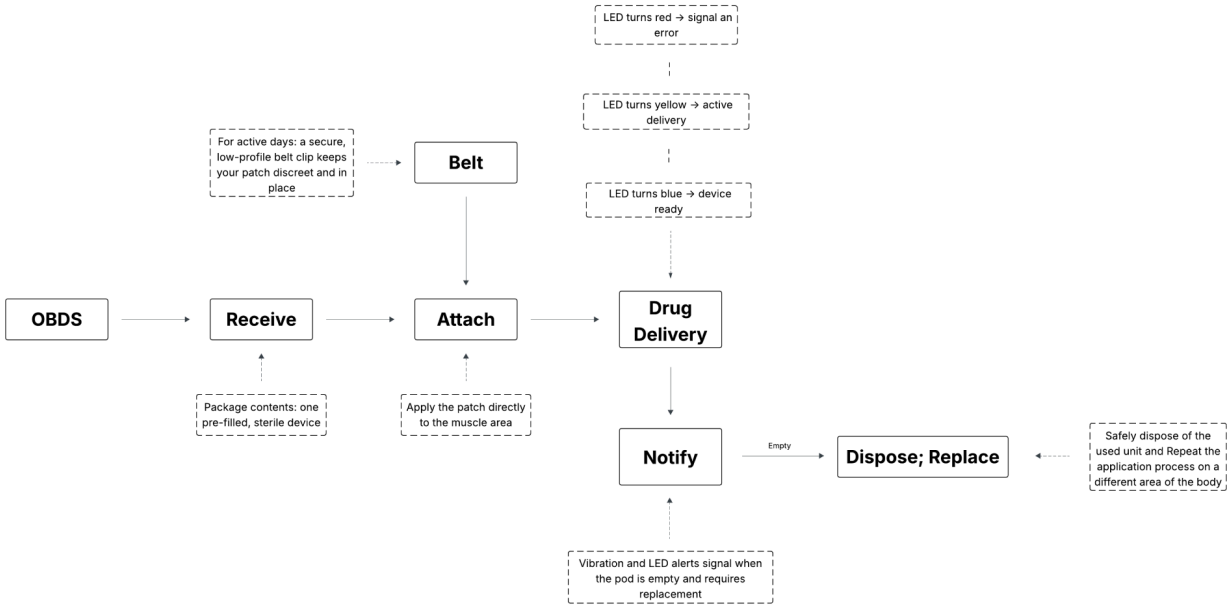
Chloe



29 years old
Wife and mother of a new born
Architect (currently on reduced hours/preparing for maternity leave)
Tech capable, but selective
Gestational Diabetes Mellitus (GDM) - 28 weeks pregnant

Chloe is a first-time expectant parent who was diagnosed with GDM during her routine second-trimester screening. Her primary focus is on the health and safety of her baby. She is already navigating a whirlwind of prenatal appointments, bodily changes, and planning for the baby's arrival. A diabetes diagnosis added an unexpected layer of complexity and anxiety to her pregnancy.

User Flow



User Flow

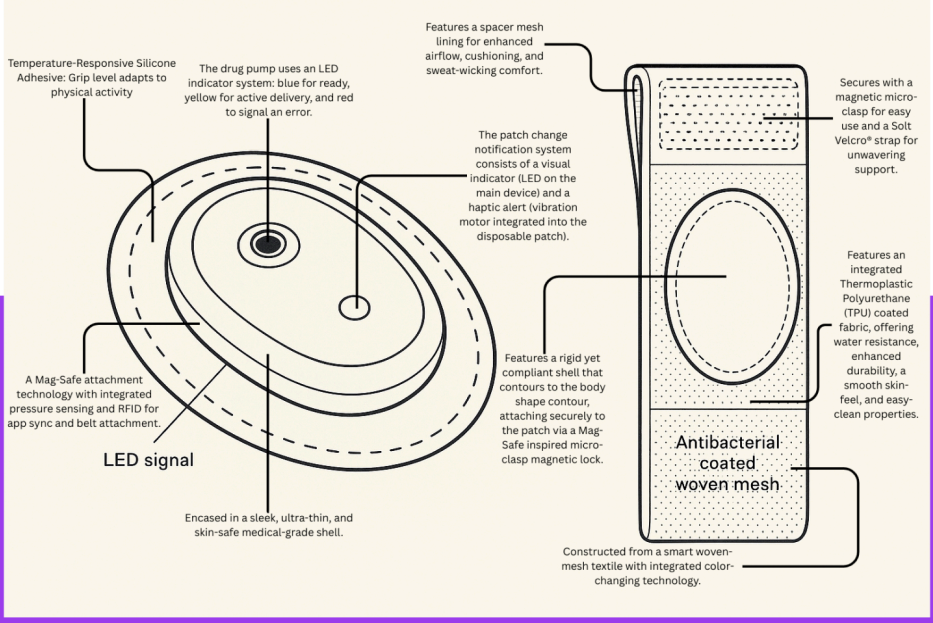
EasePatch interaction flow illustrating LED and haptic guidance through each step — from application to replacement.

Stages

- **Receive:** User gets pre-filled, sterile device
- **Attach:** Apply directly to the muscle area
- **Drug Delivery:** LED (blue-ready, yellow-active, red-error)
- **Notify:** Vibration alerts when the pod is empty
- **Dispose / Replace:** Safe removal and reapplication

Design Solution

EasePatch is a two-part wearable system that merges technology, ergonomics, and sustainability.



Device Anatomy Mock-up

Labeled schematic showing temperature-responsive adhesive, Mag-Safe microclasp, antibacterial woven mesh, and breathable TPU shell.

Core Features

- **Bio-safe design:** Medical-grade silicone and antibacterial mesh materials
- **Smart feedback:** LED and vibration system reduces uncertainty
- **Adaptive comfort:** Belt clip option for active users

- **Sustainability:** Modular and recyclable components
- **Inclusivity:** Fits various skin tones and body types

Usability Testing

EasePatch was tested with 10 participants (patients and clinicians).

Goals included evaluating comfort, clarity, and user confidence.

Key Findings

Metric	Avg. Rating (5)
Ease of Application	4.7
Comfort & Adhesion	4.6
Feedback Clarity	4.8
Discretion	4.5
Confidence	4.9

Data Analysis & Insights

Three major themes emerged from feedback:

- **Adhesion:** Desire for breathable, secure grip suitable for movement
- **Discretion:** Preference for neutral tones and low-profile design
- **Feedback Clarity:** LED and vibration cues reduced anxiety by confirming correct usage

> “I liked knowing it was working — the light and vibration gave me peace of mind.”
>
> – Participant during field test

Redesign Iterations

Based on usability data:

- Simplified LED feedback sequence (Blue = Ready, Yellow = Active, Red = Error)
- Softer, temperature-adaptive adhesives for comfort

- Updated belt clasp for adjustable fit and reduced slippage
- Modular battery component for easy replacement
- Companion app prototype for dose tracking and adherence insights

Roadmap & Feasibility

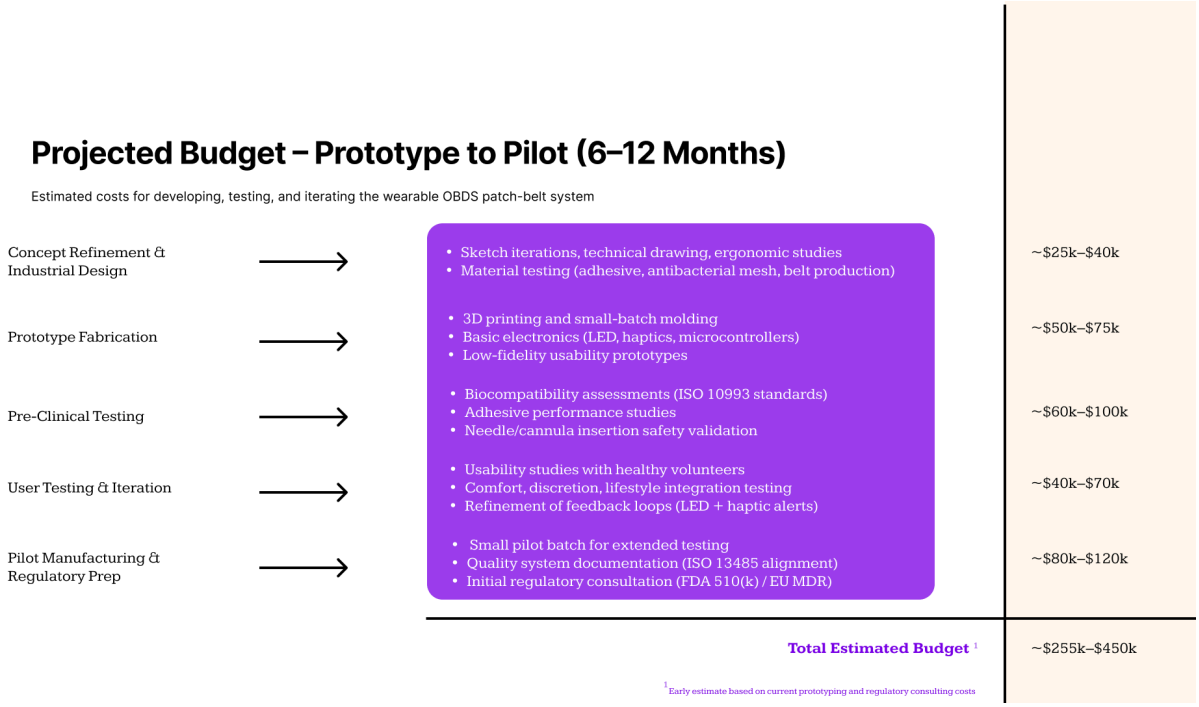
Roadmap

12-month roadmap aligning prototyping, testing, and pre-regulatory preparation milestones.

Timeline Overview

- **Months 1–3:** Concept prototyping (form factor, LED/haptic testing)
- **Months 4–6:** User testing and adhesive refinement
- **Months 7–9:** Integrated device testing with higher-fidelity prototypes
- **Months 10–12:** Final prototype and pre-regulatory preparation

Projected Budget



Projected 6–12 month development budget estimating total costs between $255K and $450K for prototype-to-pilot production.

Budget Breakdown

Stage	Cost Estimate
Concept Refinement & Industrial Design	$25K–$40K
Prototype Fabrication	$50K–$75K
Pre-Clinical Testing	$60K–$100K
User Testing & Iteration	$40K–$70K
Pilot Manufacturing & Regulatory Prep	$80K–$120K

Marketing & Positioning

EasePatch positions itself as a lifestyle-friendly healthcare device.

Marketing strategy focuses on trust, comfort, and design empathy.

Approach

- Partner with patient communities and advocacy groups
- Emphasize personalization and sustainability in messaging

- Visual branding: soft palettes, approachable typography, human-centered photography

Wrap Up

EasePatch represents how design, engineering, and empathy can intersect to create medical devices that empower patients.

Through inclusive research, sustainable design, and user testing, EasePatch moves beyond treatment and becomes part of a user’s everyday life, a possibility that helps people heal with confidence and dignity.

> This project reaffirmed that human-centered design in healthcare is about reducing friction and restoring control for patients.

Reflection

In this project, I applied skills in user research and synthesis, experience flow mapping, rapid prototyping, iterative testing, and data storytelling to design a human-centered wearable device that balanced functionality with empathy. Through continuous testing and user collaboration, I learned that emotional comfort is just as vital as technical performance in healthcare design. Inclusive research not only improves usability but also ensures that solutions remain equitable across diverse patient groups. Most importantly, this project reinforced how bridging human factors and industrial design can lead to holistic innovations that empower users and transform medical experiences into moments of trust and control.